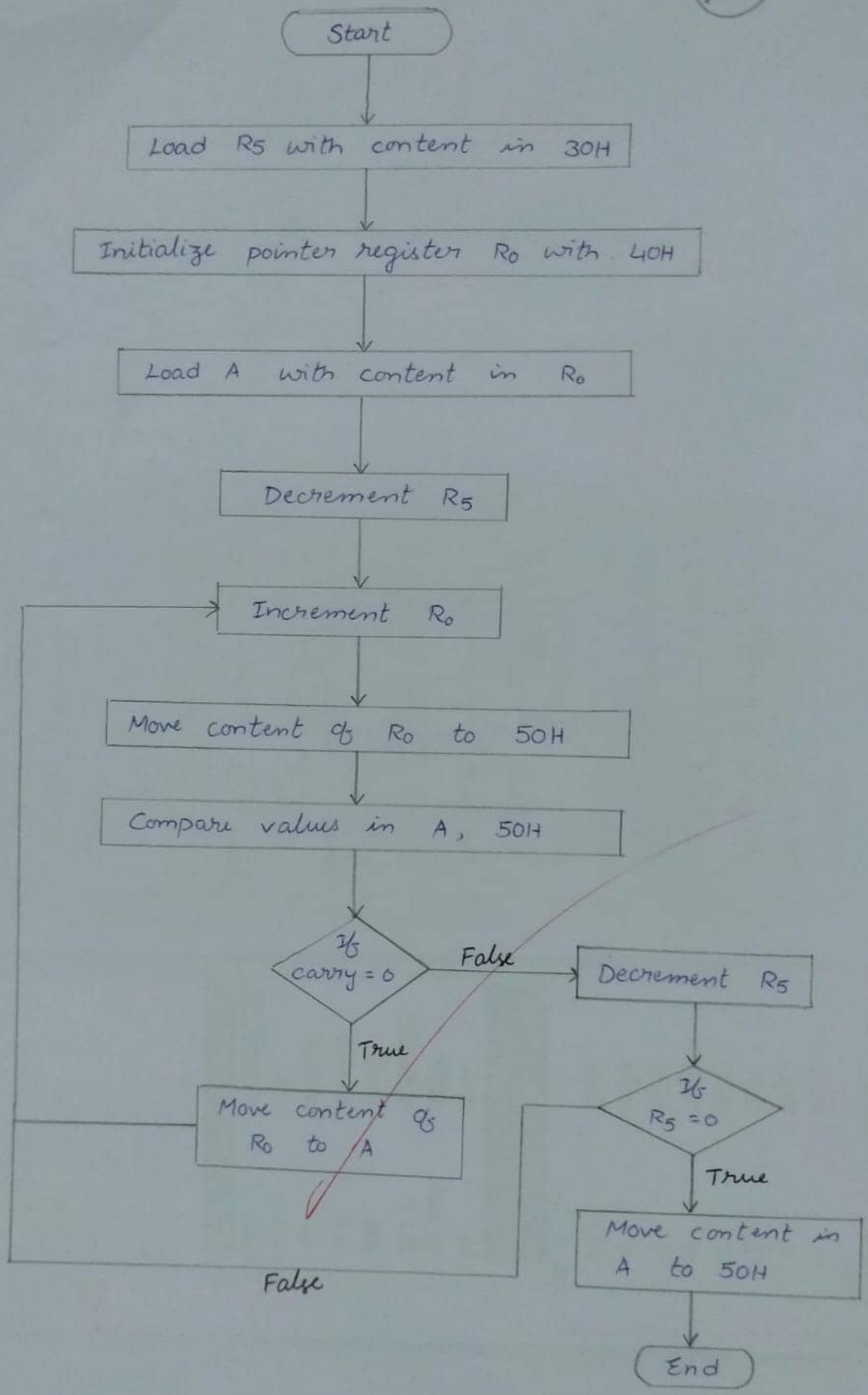


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# MODULE - II



1. AIM: 8051 assembly program to find smallest element in an array.

PROGRAM:

```
ORG 0000H
MOV R5, 30H
MOV R0, #40H
MOV A, @R0
DEC R5
L3: INC R0
MOV 50H, @R0
CJNE A, 50H, L1
SJMP L2
L1: JC L2
MOV A, @R0
L2: DJNZ R5, L3
MOV 50H, A
END
```

OUTPUT:

Address : D:0x30

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| D:0x30 | 05 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 |
| D:0x40 | 02 | 01 | 06 | 07 | 03 | 00 | 00 | 00 | 00 | 00 | 00 | 00 |
| D:0x50 | 01 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 |



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Memory 1

Address: D:0x30

```
D:0x30: 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
D:0x4B: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
D:0x66: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF
D:0x81: 07 00 00 00 00 00 00 00 00 00 00 00 00 01 00 FF 08 00 00 00 00 C7 00 00 00 00
D:0x9C: 00 00 00 00 FF 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00
D:0xB7: 00 00 00 00 00 00 00 00 00 00 00 F5 00 80 10 00 FF 00 FC 00 00 00 00 00 00 00
```

Call Stack + Locals | Memory 1

Simulation | t1: 0.00000000 sec | L:2 C:1 | CAP NUM SCRL OVR R/W

Memory 1

Address: D:0X40

```
D:0x40: 02 06 09 01 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
D:0x56: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
D:0x6C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF 07
D:0x82: 00 00 00 00 00 00 00 00 00 00 00 00 01 00 FF 08 00 00 00 00 C7 00
```

Call Stack + Locals | Memory 1

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Memory 1

Address: D:0X50

```
D:0x50: 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
D:0x66: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
D:0x7C: 00 00 00 00 FF 07 00 00 00 00 00 00 00 00 00 00 00 00 01 00 FF 08
D:0x92: 00 00 00 00 C7 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00
```

Call Stack + Locals | Memory 1

Simulation | t1: 0.00000642 sec | L:13 C:1 | CAP NUM SCRL OVR R/W

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Memory 1

Address: D:0X50

```
D:0x50: 02 06 09 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
D:0x66: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
D:0x7C: 00 00 00 00 FF 07 00 00 00 00 00 00 00 00 00 00 00 00 01 00 FF 08
D:0x92: 00 00 00 00 C7 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00
```

Call Stack + Locals | Memory 1

Memory 1

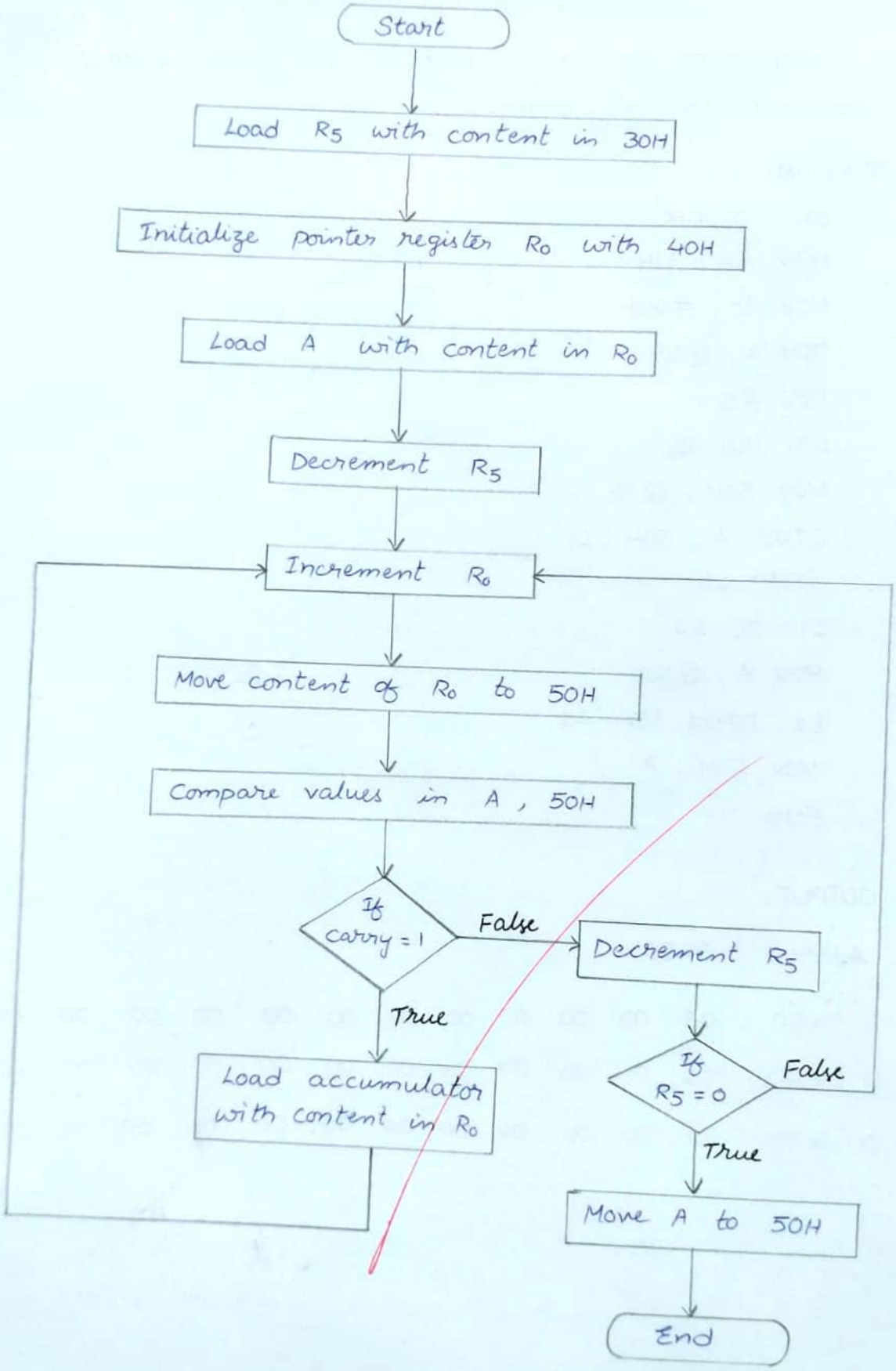
Address: D:0X50

```
D:0x50: 00 01 02 06 09 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
D:0x66: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
D:0x7C: 00 00 00 00 FF 07 00 00 00 00 00 00 00 00 00 00 00 00 01 00 FF 08
D:0x92: 00 00 00 00 C7 00 00 00 00 00 00 00 00 00 FF 00 00 00 00 00 00 00
```

Call Stack + Locals | Memory 1

Simulation

FLOWCHART :




Scanned with OKEN Scanner

2. AIM: 8051 program to find largest element in an array.

PROGRAM:

```
ORG 0000H
MOV R5, 30H
MOV R0, #40H
MOV A, @R0
DEC R5
L3: INC R0
MOV 50H, @R0
CJNE A, 50H, L1
SJMP L2
L1: JNC L2
MOV A, @R0
L2: DJNZ R5, L3
MOV 50H, A
END
```

OUTPUT:

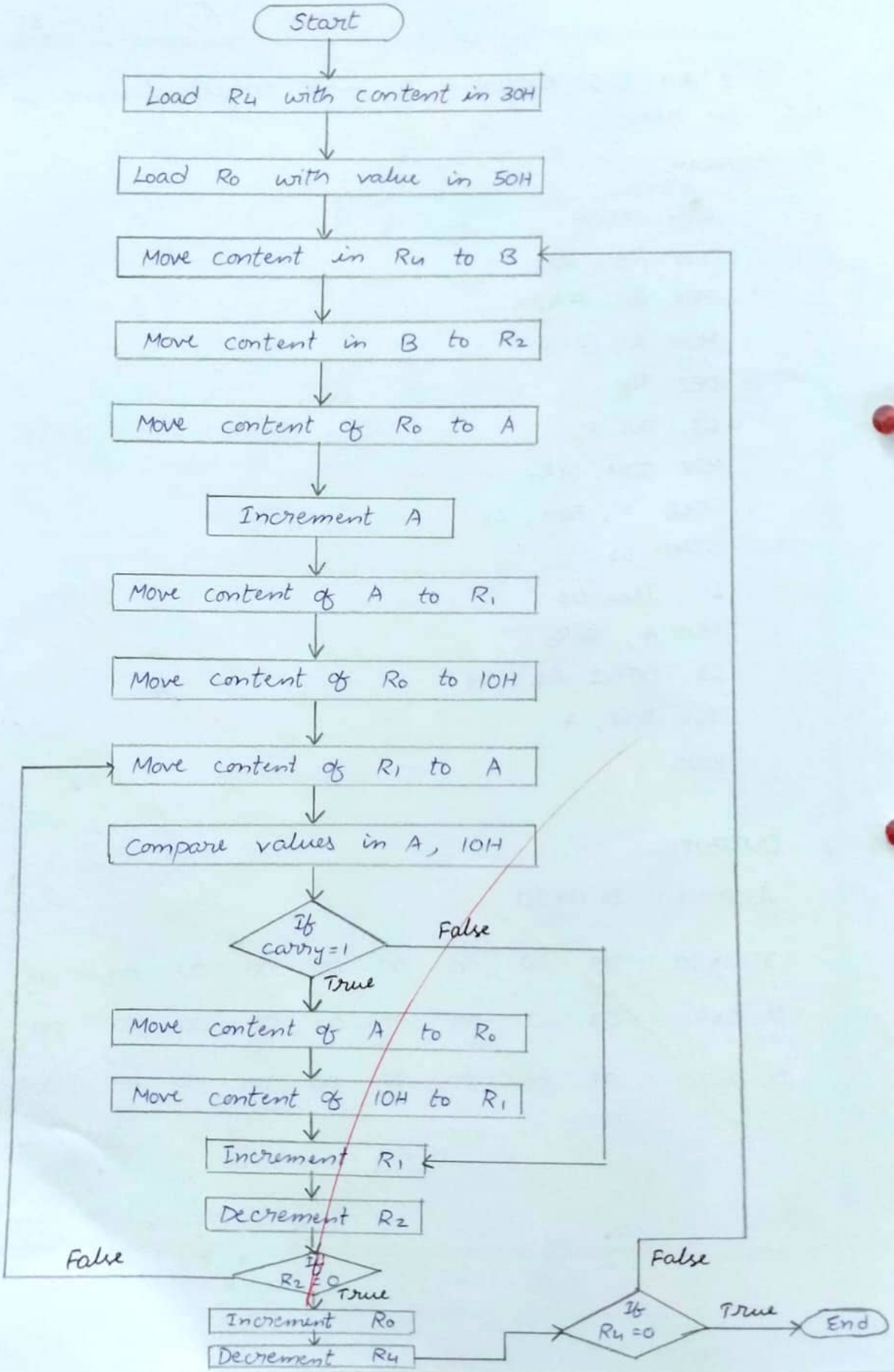
Address : D: 0x30

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| D: 0x30 | 05 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 |
| D: 0x40 | 02 | 06 | 09 | 01 | 03 | 00 | 00 | 00 | 00 | 00 |
| D: 0x50 | 09 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 |



Scanned with OKEN Scanner

FLOWCHART:




Scanned with OKEN Scanner

3. AIM: 8051 program to sort array in ascending order.

PROGRAM:

```
ORG 0000H
MOV R4, 30H
MOV R0, #50H
L1: MOV B, R4
MOV R2, B
MOV A, R0
INC A
MOV R1, A
L2: MOV 10H, @R0
MOV A, @R1
CJNE A, 10H, L3
SJMP NEXT
L3: JC EXCHANGE
SJMP NEXT
EXCHANGE: MOV @R0, A
MOV @R1, 10H
NEXT: INC R1
DJNZ R2, L2
INC R0
DJNZ R4, L1
END
```

OUTPUT:

Address: D: 0x30

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| D: 0x30 | 04 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 |
| D: 0x50 | 02 | 06 | 09 | 01 | 00 | 00 | 00 | 00 | 00 | 00 | 00 |
| D: 0x50 | 00 | 01 | 02 | 06 | 09 | | | | | | |



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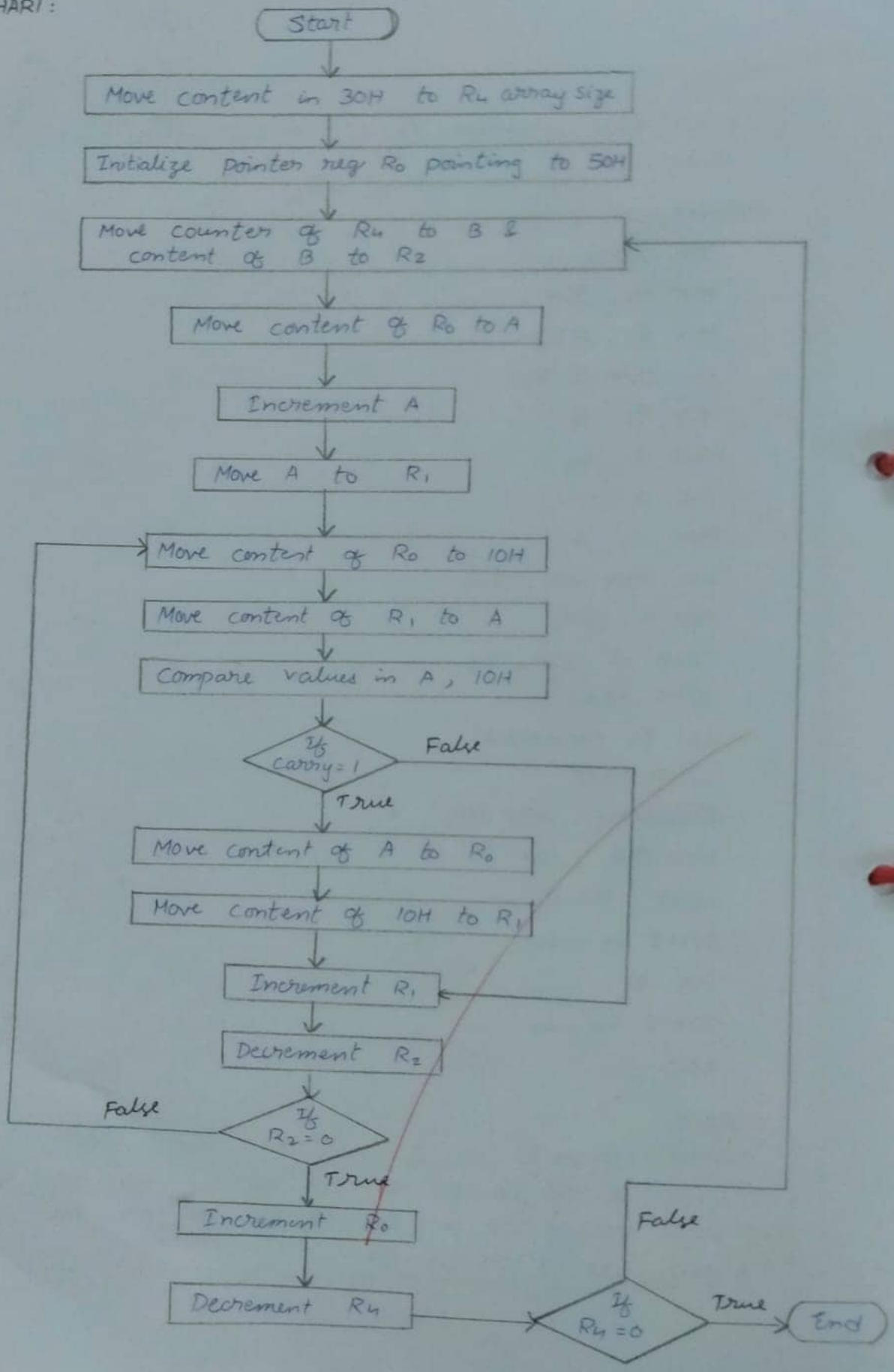
FLOWCHART:




Scanned with OKEN Scanner

4. AIM: 8051 program to sort array in descending order.

PROGRAM:

```
ORG 0000H
MOV R0, #50H
MOV R4, 30H
L1: MOV B, R4
MOV R2, B
MOV A, R0
INC A
MOV R1, A
L2: MOV 10H, @R0
MOV A, @R1
CJNE A, 10H, L3
SJMP NEXT
L3: JNC EXCHANGE
SJMP NEXT
EXCHANGE: MOV @R0, A
MOV @R1, 10H
NEXT: INC R1
DJNZ: R2, L2
INC R0
DJNZ R4, L1
END
```

OUTPUT:

Address : D:0x30

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| D:0x30 | 05 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 |
| D:0x50 | 04 | 05 | 24 | 19 | 12 | 00 | 00 | 00 | 00 | 00 | 00 |
| D:0x50 | 24 | 19 | 12 | 05 | 04 | 00 | 00 | 00 | 00 | 00 | 00 |



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